EXPERIMENT 1

Generate Text Embedding and Perform Semantic Similarity Search

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AIM

To convert text into numerical vectors and find the most similar document using cosine similarity.

PROCEDURE

1. Import TfidfVectorizer and cosine_similarity from scikit-learn.
2. Create sample documents and a search query.
3. Convert the documents and query into TF-IDF vectors.
4. Calculate cosine similarity between the query and each document.
5. Display similarity scores and the most similar document.

PROGRAM

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

documents = [
    "Python is a programming language.",
    "Machine learning uses data to make predictions.",
    "Artificial intelligence makes machines intelligent."
]

query = "Tell me about machine learning"

vectorizer = TfidfVectorizer()
doc_vectors = vectorizer.fit_transform(documents)
query_vector = vectorizer.transform([query])

scores = cosine_similarity(query_vector, doc_vectors)[0]

print("Search Query:", query)
print("\nSimilarity Scores:")
for i, score in enumerate(scores):
    print("Document", i + 1, ":", round(score, 3))

best_index = scores.argmax()
print("\nMost Similar Document:")
print(documents[best_index])
```

OUTPUT

```
similarity = cosine_similarity(query_embedding, document_embeddings)

print("Query:", query)
print("\nSimilarity Results:")

for i, score in enumerate(similarity[0]):
    print(documents[i], ":", round(score, 3))

best_match = similarity.argmax()
print("\nMost Similar Document:", documents[best_match])
```

```
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.13/dist-packages (1.6.1)
Requirement already satisfied: numpy>=1.19.5 in /usr/local/lib/python3.13/dist-packages (from scikit-learn) (
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.13/dist-packages (from scikit-learn) (1
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.13/dist-packages (from scikit-learn) (
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.13/dist-packages (from scikit-l
Requirement already satisfied: cloudpickle>=3.0 in /usr/local/lib/python3.13/dist-packages (from joblib>=1.2.
Query: Tell me about artificial intelligence

Similarity Results:
Python is a programming language : 0.0
Java is used for software development : 0.0
Machine learning is a branch of artificial intelligence : 0.576
Deep learning uses neural networks : 0.0

Most Similar Document: Machine learning is a branch of artificial intelligence
```

RESULT

Thus, text documents were converted into vectors and the most similar document was retrieved using cosine similarity.